Installing electric stove in an Amel Super Maramu

On Lady Annila (SM 232) we have had an ENO stove, propane driven, that have served us well over the years. When it was time to change we started to think about leaving propane and going electric. There are different options for doing this and also the question if one want to make adjustments to the space designated for the stove. On the Super Maramu this space is less wide than on Super Maramu 2000 with the big Eno stove.

After going through the market, actors offering electric stoves, we soon found that combining our demands to the market offering met some problems. We where looking for Induction cook top and a convection oven. We found that this combination was hard to find but changing to ceramic cooking top changed the situation. But, been cooking food on a propane stove going over to ceramic cooking top was not the preferred choice.

So, we decided to design a stainless steel box that would contain both the cooking top and the convection oven with a design to evacuate hot air aft of the unit and see to that the fresh air intake was less compromised by hot air. The size of the box is slightly wider and higher than the old ENO stove.

In parallel we started to look for separate units for cooking top and convection oven. Induction cooking top are getting more common and our choice at the end was Kenyon Silken 2 induction cook top (www.cookingwithkenyon.com) design for out door kitchen use. To find a convection oven was far harder, our choice at the end was a unit from German producer Steba, KB E350, a table top model that was very well insulated (www.steba.com).

What we looked for outside the above demands was that the units worked with AC 220v/60Hz, the European standard, energy efficiency, and well insulated oven to avoid unnecessary heat going outside the unit and of course fitted in the stainless steel box.

What about power demands?
The stoves maximum consumption is 3400W which made us choose an inverter providing 4000W. We decided to upgrade our charging system and solar panels to latest version. The charging system is delivered by Swiss producer Studer and the solar panels from Sun Power (366w/panel, 96 cell, 60v and efficiency around 22-23%) with the moderate size of 1m*1.5m. In total we have 3 panels on our arch and 2 panels on the fixed roof over the cockpit.

We decided to wait updating our battery bank so the present bank of 8 Trojan 6v 225Ah Golf Cart batteries would serve over the first season.

Put it in place
Installing the new stove was a plug and play process. The propane connection was disconnected, the old wall mat was taken away and a thins metal film was glued in place replacing the mat. On the sides small squares where glued to protect the Mahogany sides from over heating. Using the same holes for hanging the unit worked perfectly.

Our experience after 2 months sailing and only being on anchor
The change has worked far better than expected, much less heat in the kitchen (due to induction cook top and well insulated oven), far less power consumption than expected and last but not least the greasy surface that always was the case around the old stove is gone.

What about power consumption, we used our diesel generator for a total of 1.5 hours during this test period meaning all power came from the solar panels and the old battery bank. Did we do any change in our cooking habits, the only one was that we started cooking dinner a little earlier to make us of the solar panels when cooking.

Would we recommend this change, absolutely. We found the Steba oven working very well and did not experience any problems with warm sides. This was tested during a cooking session over in total 2 hours use of both cooking top and oven.

Do you have to have as much solar panel capacity as we have? No, you have your diesel generator (Onan) that happily will provide you with the needed power. We use the extra power to manage power to our warm water heater and water maker which leave us to run the Onan when we use the washing machine or the dishwasher.